# Data Analysis
# AMS315

## Chapter 11

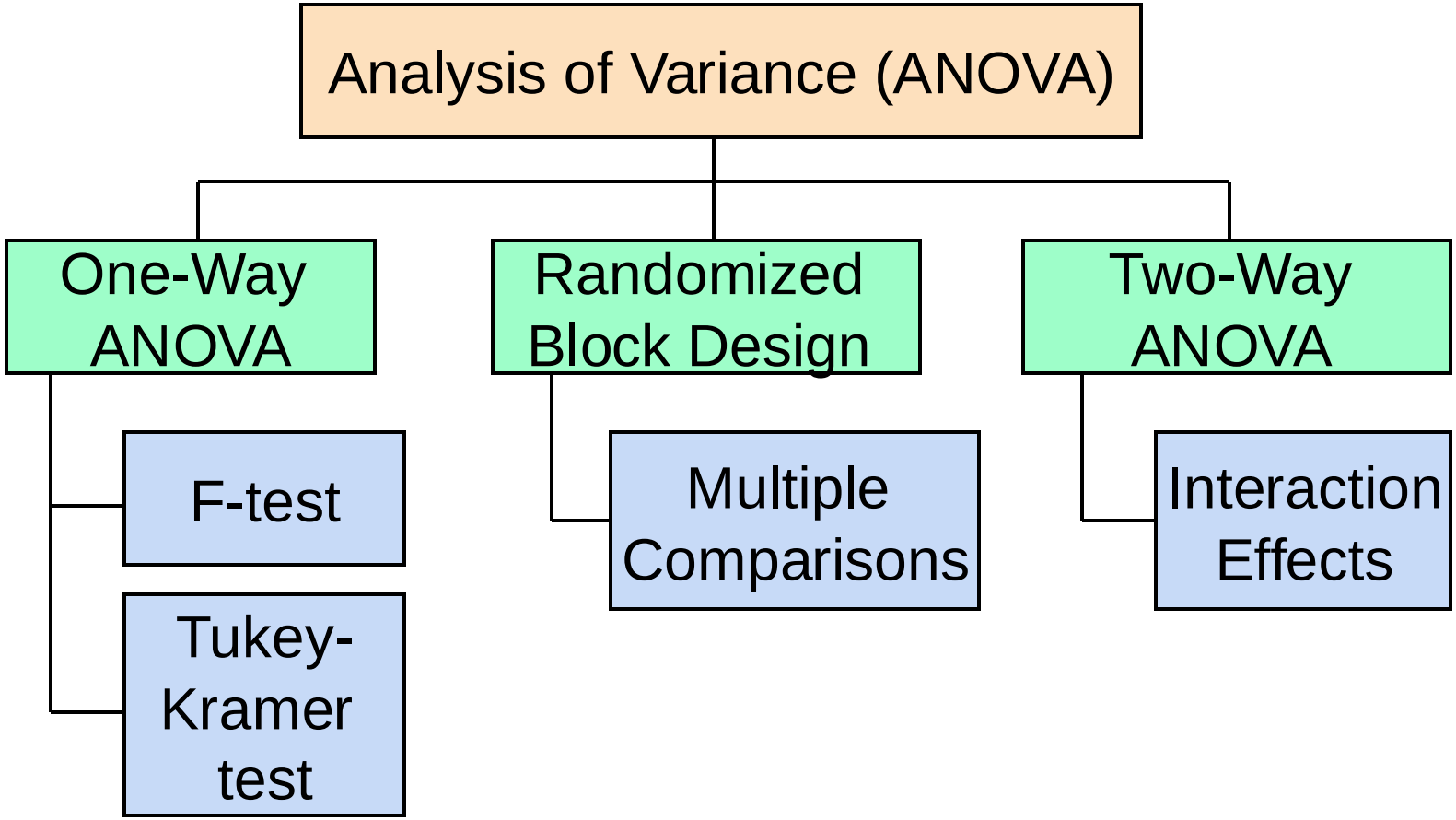
Analysis of Variance

# Learning Objectives

**In this chapter, you learn:**

- The basic concepts of experimental design
- How to use one-way analysis of variance to test for differences among the means of several populations (also referred to as “groups” in this chapter)
- When to use a randomized block design
- How to use two-way analysis of variance and the concept of interaction

# Chapter Overview



# General ANOVA Setting

- Investigator controls one or more independent variables
  - Called factors (or treatment variables)
  - Each factor contains two or more levels (or groups or categories/classifications)
- Observe effects on the dependent variable
  - Response to levels of independent variable
- Experimental design: the plan used to collect the data

# Completely Randomized Design

- Experimental units (subjects) are assigned randomly to treatments
  - Subjects are assumed homogeneous
- Only one factor or independent variable
  - With two or more treatment levels
- Analyzed by one-way analysis of variance (ANOVA)

# One-Way Analysis of Variance

- Evaluate the difference among the means of three or more groups

| | |
|---|---|
| Examples: | Accident rates for 1st, 2nd, and 3rd shift |
| | Expected mileage for five brands of tires |

- Assumptions
  - Populations are normally distributed
  - Populations have equal variances
  - Samples are randomly and independently drawn

# Hypotheses of One-Way ANOVA

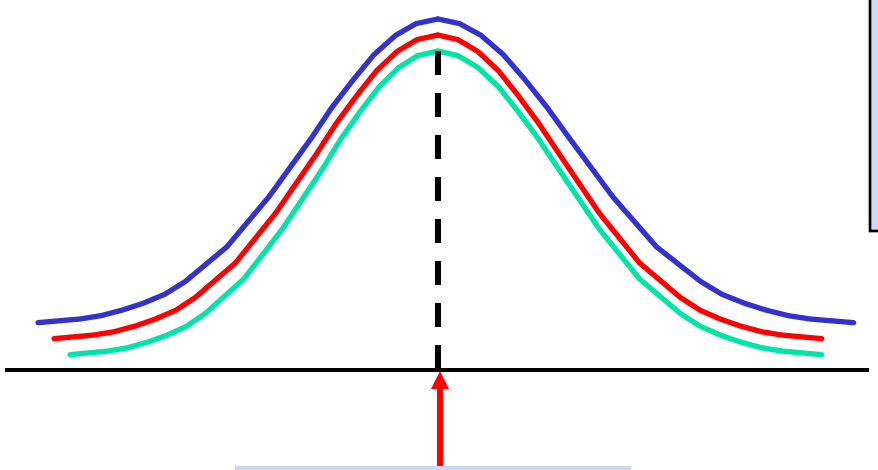
- $H_0: \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_c$
  - All population means are equal
  - i.e., no treatment effect (no variation in means among groups)

- $H_1$: Not all of the population means are the same
  - At least one population mean is different
  - i.e., there is a treatment effect
  - Does not mean that all population means are different (some pairs may be the same)

# One-Factor ANOVA

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_c$$

$$H_1 : \text{Not all } \mu_j \text{ are the same}$$



$$\mu_1 = \mu_2 = \mu_3$$

All Means are the same:
The Null Hypothesis is True
(No Treatment Effect)

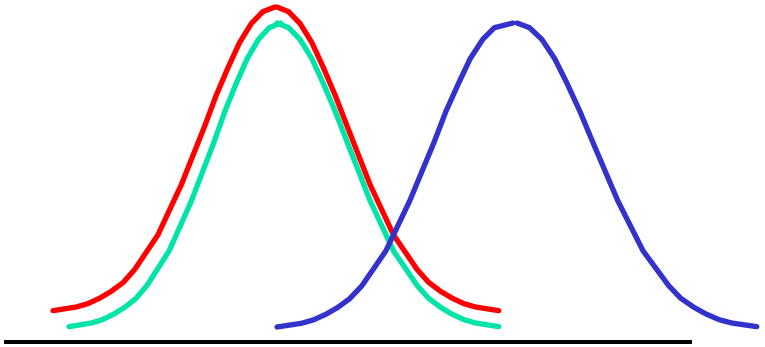
# One-Factor ANOVA

*(continued)*

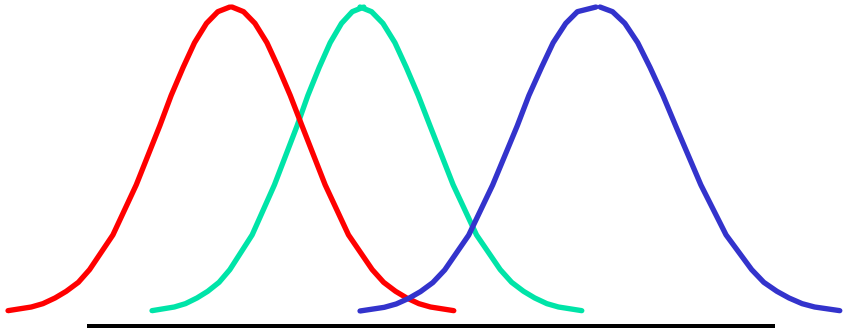
$$H_0 : \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_c$$

$$H_1 : \text{Not all } \mu_j \text{ are the same}$$

At least one mean is different:
The Null Hypothesis is NOT true
(Treatment Effect is present)



or



$$\mu_1 = \mu_2 \neq \mu_3$$

$$\mu_1 \neq \mu_2 \neq \mu_3$$

# Partitioning the Variation

- Total variation can be split into two parts:

$$SST = SSA + SSW$$

SST = Total Sum of Squares
(Total variation)
SSA = Sum of Squares Among Groups
(Among-group variation)
SSW = Sum of Squares Within Groups
(Within-group variation)

# Partitioning the Variation

*(continued)*

$$SST = SSA + SSW$$

Total Variation = the aggregate dispersion of the individual data values across the various factor levels (SST)

Among-Group Variation = dispersion between the factor sample means (SSA)

Within-Group Variation = dispersion that exists among the data values within a particular factor level (SSW)

# Partition of Total Variation

**Total Variation (SST)**

d.f. = n – 1

= **Variation Due to Factor (SSA)** + **Variation Due to Random Sampling (SSW)**

d.f. = c – 1

d.f. = n – c

Commonly referred to as:

- Sum of Squares Between
- Sum of Squares Among
- Sum of Squares Explained
- Among Groups Variation

Commonly referred to as:

- Sum of Squares Within
- Sum of Squares Error
- Sum of Squares Unexplained
- Within-Group Variation

# Total Sum of Squares

$$\boxed{SST} = SSA + SSW$$

$$\boxed{SST = \sum_{j=1}^{c}\sum_{i=1}^{n_j}(X_{ij} - \bar{\bar{X}})^2}$$

Where:

SST = Total sum of squares

c = number of groups (levels or treatments)

$n_j$ = number of observations in group j

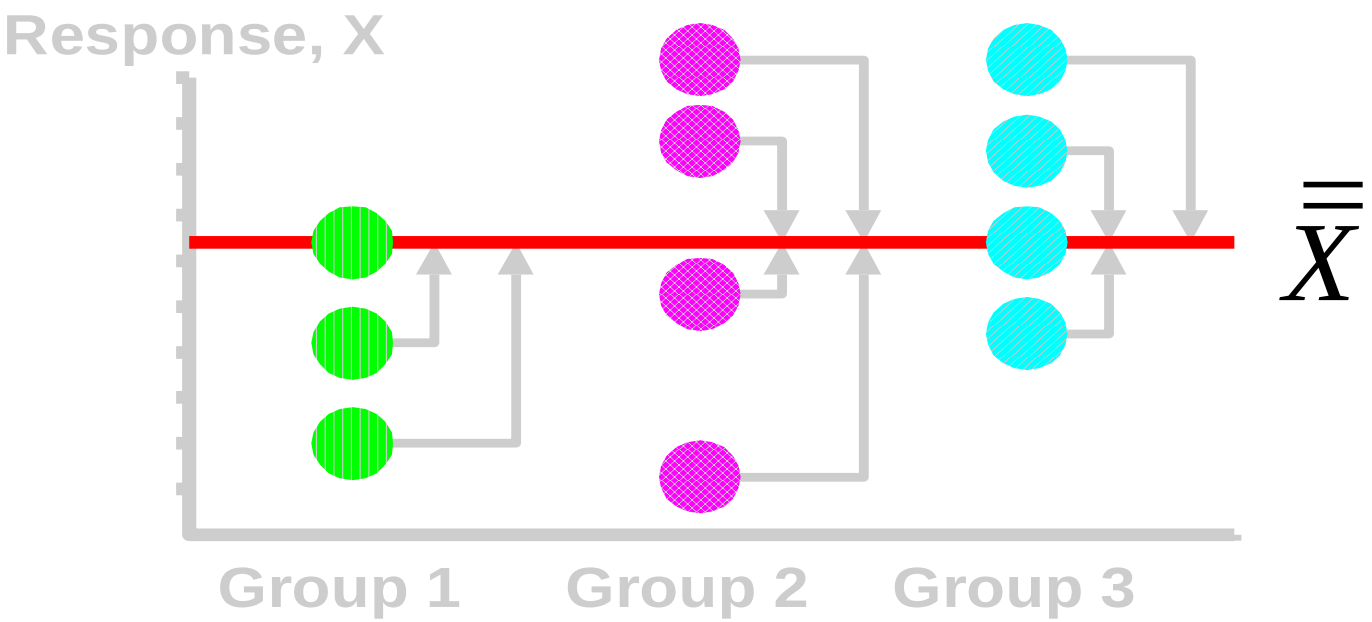
$X_{ij}$ = $i^{th}$ observation from group j

$\bar{\bar{X}}$ = grand mean (mean of all data values)

# Total Variation

*(continued)*

$$SST = (X_{11} - \bar{\bar{X}})^2 + (X_{12} - \bar{\bar{X}})^2 + \ldots + (X_{cn_c} - \bar{\bar{X}})^2$$



# Among-Group Variation

$$SST = \boxed{SSA} + SSW$$

$$\boxed{SSA = \sum_{j=1}^{c} n_j(\bar{X}_j - \bar{\bar{X}})^2}$$

Where:

SSA = Sum of squares among groups

c = number of groups

$n_j$ = sample size from group j

$\bar{X}_j$ = sample mean from group j

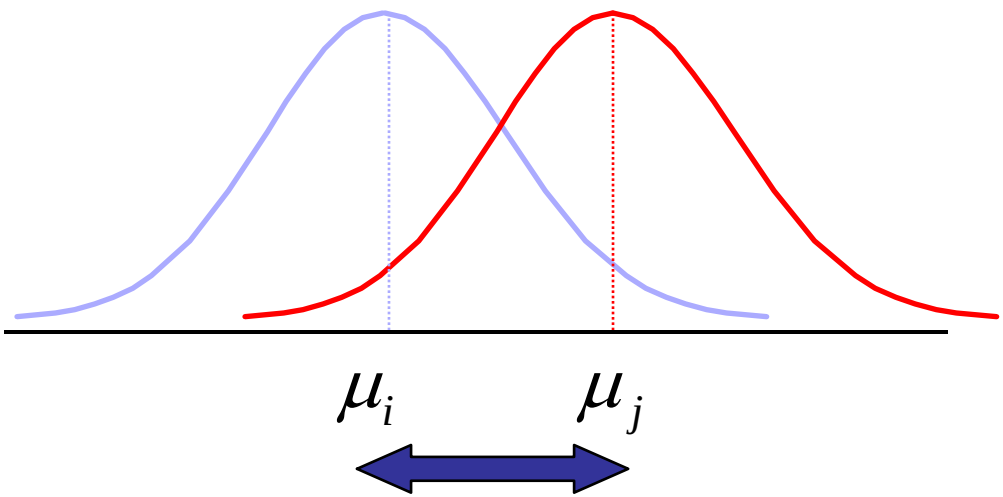
$\bar{\bar{X}}$ = grand mean (mean of all data values)

# Among-Group Variation

*(continued)*

$$SSA = \sum_{j=1}^{c} n_j(\bar{X}_j - \bar{\bar{X}})^2$$

Variation Due to
Differences Among Groups



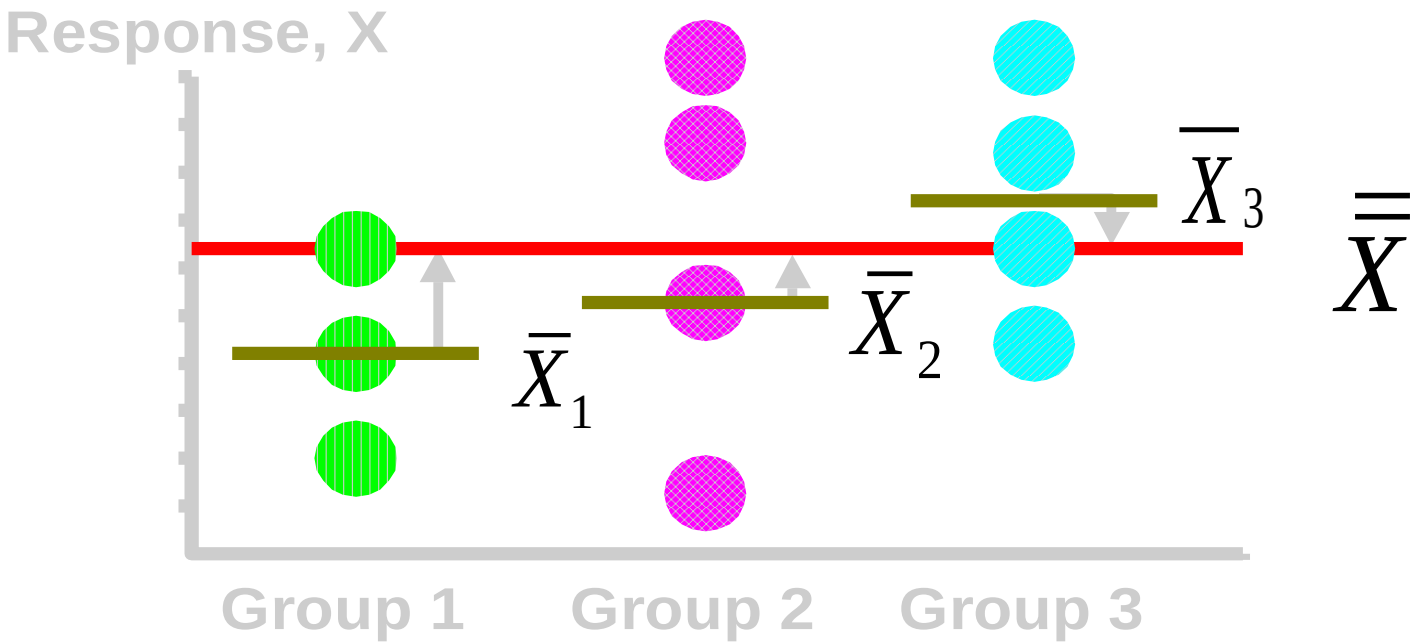

$$MSA = \frac{SSA}{c-1}$$

Mean Square Among =
SSA/degrees of freedom

# Among-Group Variation

*(continued)*

$$SSA = n_1(\bar{x}_1 - \bar{\bar{x}})^2 + n_2(\bar{x}_2 - \bar{\bar{x}})^2 + \ldots + n_c(\bar{x}_c - \bar{\bar{x}})^2$$



# Within-Group Variation

$$SST = SSA + \boxed{SSW}$$

$$\boxed{SSW = \sum_{j=1}^{c} \sum_{i=1}^{n_j} (X_{ij} - \bar{X}_j)^2}$$

Where:

SSW = Sum of squares within groups

c = number of groups

$n_j$ = sample size from group j

$\bar{X}_j$ = sample mean from group j

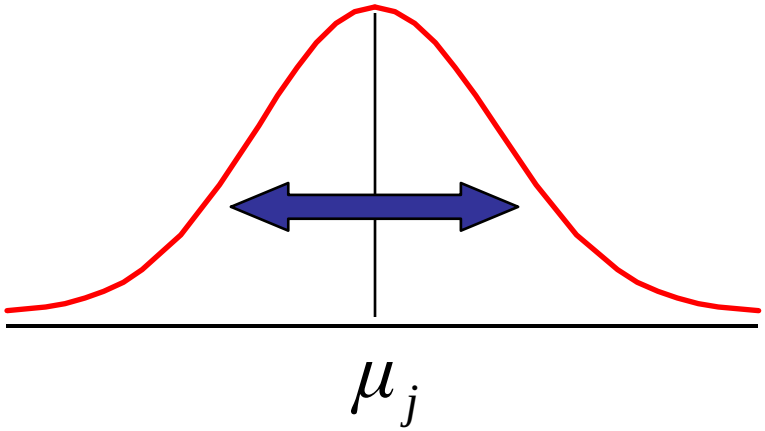
$X_{ij}$ = $i^{th}$ observation in group j

# Within-Group Variation

*(continued)*

$$SSW = \sum_{j=1}^{c} \sum_{i=1}^{n_j} (X_{ij} - \overline{X}_j)^2$$

Summing the variation within each group and then adding over all groups



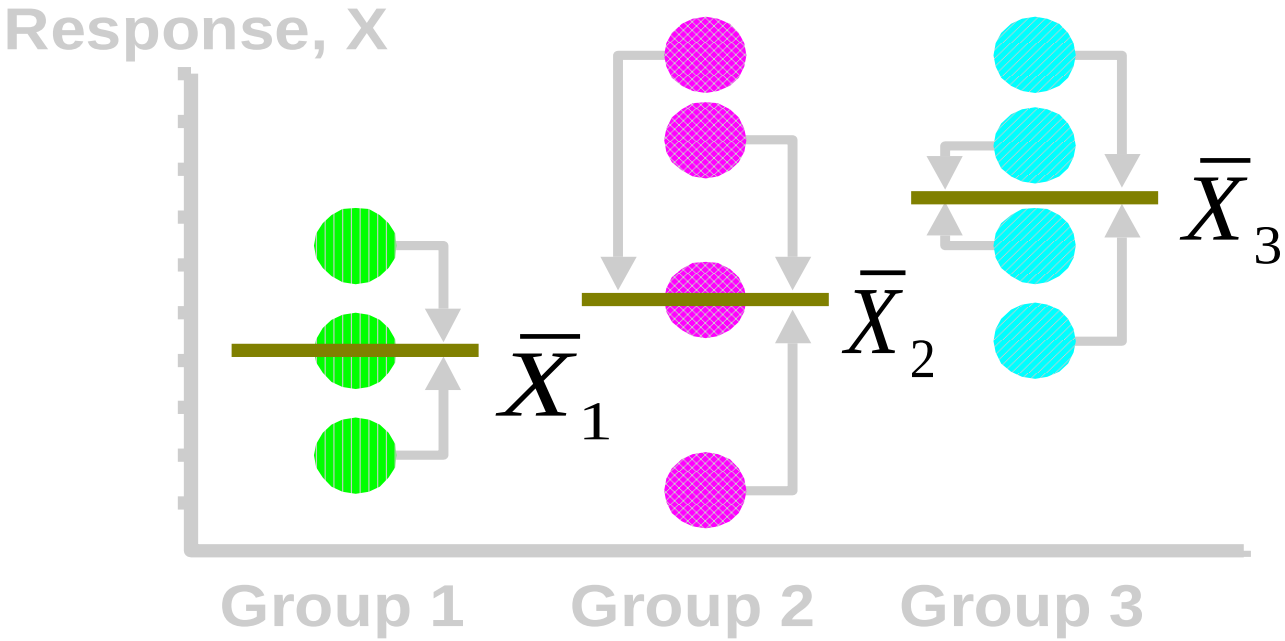

$$MSW = \frac{SSW}{n - c}$$

Mean Square Within = SSW/degrees of freedom

# Within-Group Variation

*(continued)*

$$SSW = (x_{11} - \overline{X}_1)^2 + (X_{12} - \overline{X}_2)^2 + \ldots + (X_{cn_c} - \overline{X}_c)^2$$



# Obtaining the Mean Squares

$$MSA = \frac{SSA}{c-1}$$

$$MSW = \frac{SSW}{n-c}$$

$$MST = \frac{SST}{n-1}$$

# One-Way ANOVA Table

| Source of Variation | SS | df | MS (Variance) | F ratio |
|---|---|---|---|---|
| Among Groups | SSA | c - 1 | $MSA = \frac{SSA}{c - 1}$ | $F = \frac{MSA}{MSW}$ |
| Within Groups | SSW | n - c | $MSW = \frac{SSW}{n - c}$ | |
| Total | SST = SSA+SSW | n - 1 | $MST = \frac{SST}{n - 1}$ | |

c = number of groups

n = sum of the sample sizes from all groups

df = degrees of freedom

# One-Way ANOVA F Test Statistic

$H_0$: $\mu_1 = \mu_2 = \ldots = \mu_c$

$H_1$: At least two population means are different

- Test statistic

$$F = \frac{MSA}{MSW}$$

*MSA* is mean squares among groups

*MSW* is mean squares within groups

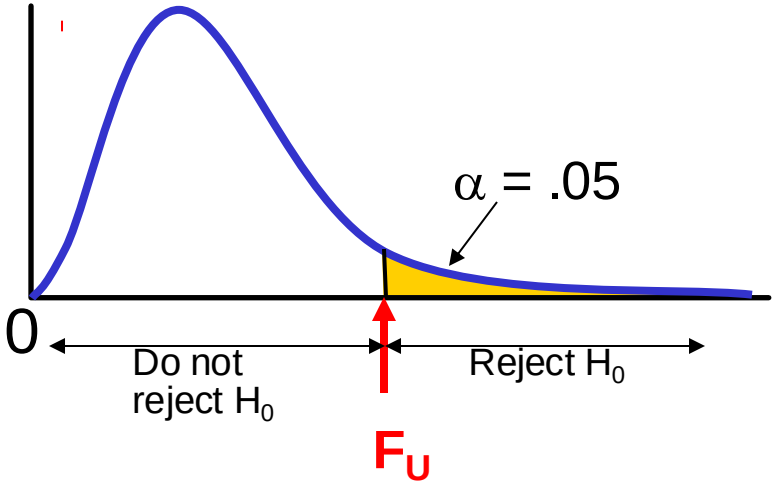
- Degrees of freedom
  - $df_1 = c - 1$ (c = number of groups)
  - $df_2 = n - c$ (n = sum of sample sizes from all populations)

# Interpreting One-Way ANOVA F Statistic

- The F statistic is the ratio of the among estimate of variance and the within estimate of variance
  - The ratio must always be positive
  - $df_1 = c - 1$ will typically be small
  - $df_2 = n - c$ will typically be large

**Decision Rule:**

- Reject $H_0$ if $F > F_U$, otherwise do not reject $H_0$



# One-Way ANOVA F Test Example

You want to see if three different golf clubs yield different distances. You randomly select five measurements from trials on an automated driving machine for each club. At the 0.05 significance level, is there a difference in mean distance?

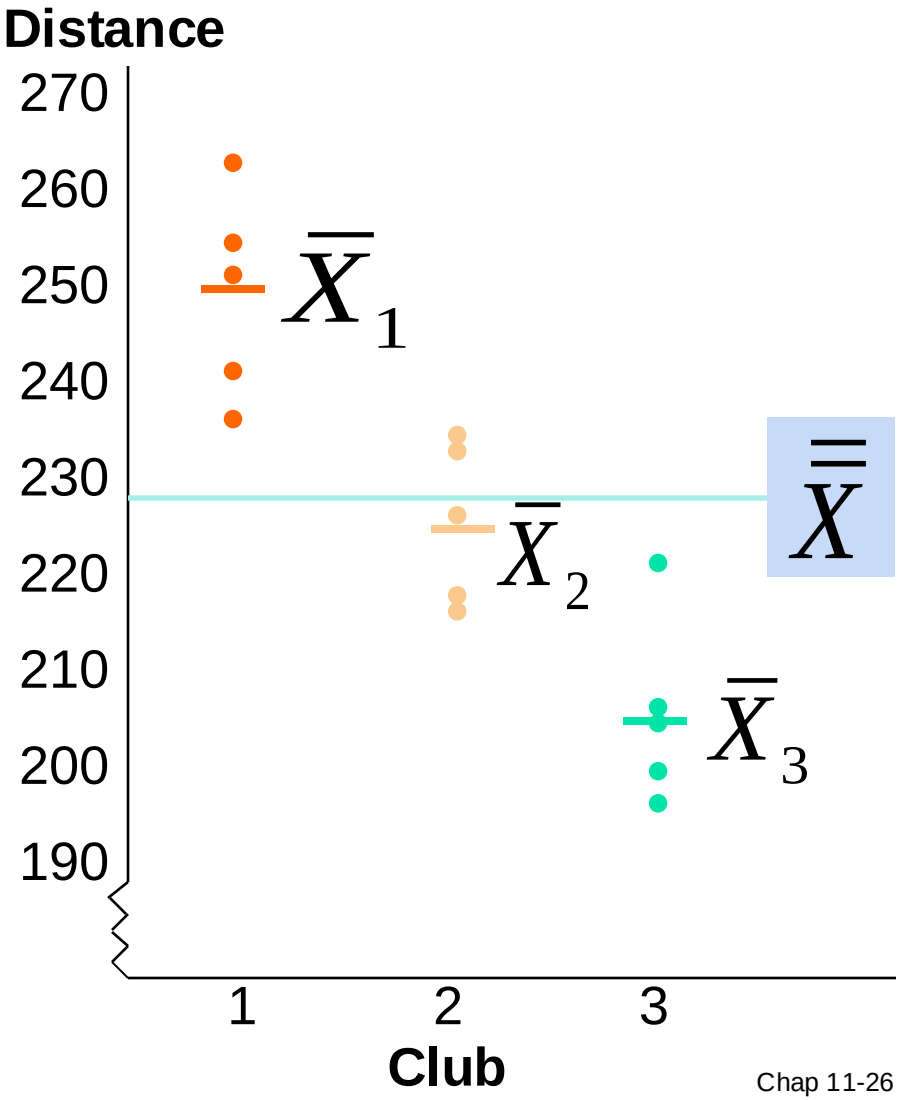
| Club 1 | Club 2 | Club 3 |
|---|---|---|
| 254 | 234 | 200 |
| 263 | 218 | 222 |
| 241 | 235 | 197 |
| 237 | 227 | 206 |
| 251 | 216 | 204 |

# One-Way ANOVA Example: Scatter Diagram

| Club 1 | Club 2 | Club 3 |
|---|---|---|
| 254 | 234 | 200 |
| 263 | 218 | 222 |
| 241 | 235 | 197 |
| 237 | 227 | 206 |
| 251 | 216 | 204 |

| | | |
|---|---|---|
| $\bar{x}_1 = 249.2$ | $\bar{x}_2 = 226.0$ | $\bar{x}_3 = 205.8$ |

$\bar{\bar{x}} = 227.0$



# One-Way ANOVA Example Computations

| Club 1 | Club 2 | Club 3 |
|---|---|---|
| 254 | 234 | 200 |
| 263 | 218 | 222 |
| 241 | 235 | 197 |
| 237 | 227 | 206 |
| 251 | 216 | 204 |

$\bar{X}_1 = 249.2$ $\quad n_1 = 5$

$\bar{X}_2 = 226.0$ $\quad n_2 = 5$

$\bar{X}_3 = 205.8$ $\quad n_3 = 5$

$\bar{\bar{X}} = 227.0$

$n = 15$

$c = 3$

$$SSA = 5\,(249.2 - 227)^2 + 5\,(226 - 227)^2 + 5\,(205.8 - 227)^2 = 4716.4$$

$$SSW = (254 - 249.2)^2 + (263 - 249.2)^2 + \ldots + (204 - 205.8)^2 = 1119.6$$

$$MSA = 4716.4 / (3-1) = 2358.2$$

$$MSW = 1119.6 / (15-3) = 93.3$$

$$F = \frac{2358.2}{93.3} = 25.275$$

# One-Way ANOVA Example Solution

$H_0$: $\mu_1 = \mu_2 = \mu_3$

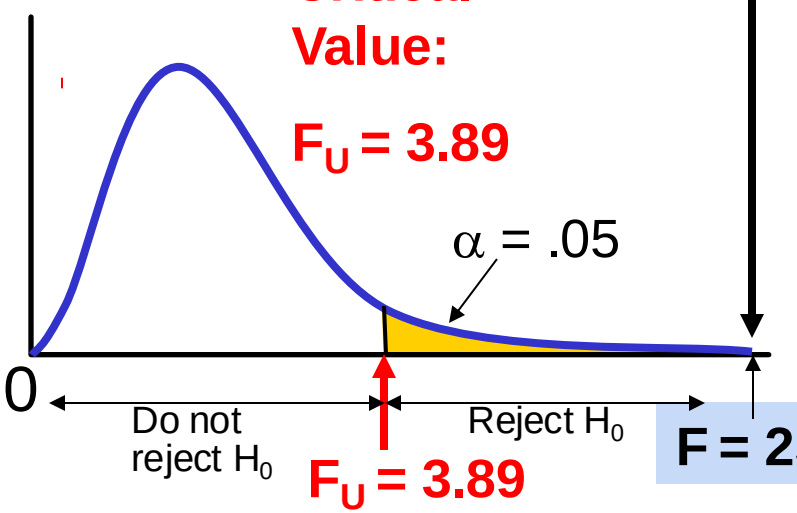
$H_1$: $\mu_j$ not all equal

$\alpha = 0.05$

$df_1 = 2$ $df_2 = 12$

**Critical Value:**

$F_U = 3.89$




**F = 25.275**

**Test Statistic:**

$$F = \frac{MSA}{MSW} = \frac{2358.2}{93.3} = 25.275$$

**Decision:**

Reject $H_0$ at $\alpha = 0.05$

**Conclusion:**

There is evidence that at least one $\mu_j$ differs from the rest

# One-Way ANOVA Excel Output

EXCEL: tools | data analysis | ANOVA: single factor

| **SUMMARY** | | | | | | |
|---|---|---|---|---|---|---|
| *Groups* | *Count* | *Sum* | *Average* | *Variance* | | |
| Club 1 | 5 | 1246 | 249.2 | 108.2 | | |
| Club 2 | 5 | 1130 | 226 | 77.5 | | |
| Club 3 | 5 | 1029 | 205.8 | 94.2 | | |
| **ANOVA** | | | | | | |
| *Source of Variation* | *SS* | *df* | *MS* | *F* | *P-value* | *F crit* |
| Between Groups | **4716.4** | **2** | **2358.2** | **25.275** | 4.99E-05 | **3.89** |
| Within Groups | **1119.6** | **12** | **93.3** | | | |
| Total | 5836.0 | 14 | | | | |

# Class Exercise

## Golf Club Distance Test

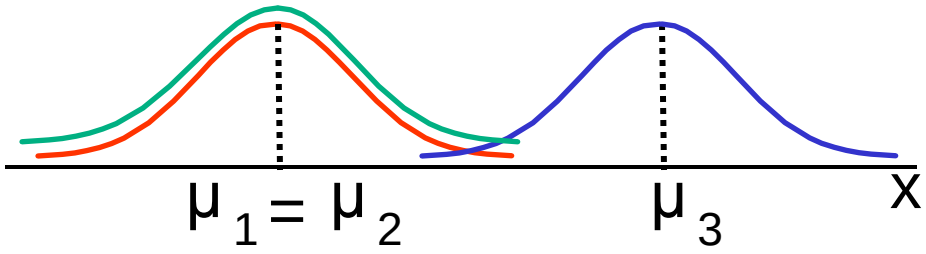
You want to test if three different golf clubs produce significantly different mean driving distances. The recorded distances (in meter) for five trials each are as follows:

| Club 1 | Club 2 | Club 3 |
|---|---|---|
| 254 | 234 | 200 |
| 263 | 218 | 222 |
| 241 | 235 | 197 |
| 237 | 227 | 206 |
| 251 | 216 | 204 |

At the $\alpha = 0.05$ significance level, is there evidence to suggest a difference in mean driving distances?

# The Tukey-Kramer Procedure

- Tells which population means are significantly different
  - e.g.: $\mu_1 = \mu_2 \neq \mu_3$
  - Done after rejection of equal means in ANOVA
- Allows pair-wise comparisons
  - Compare absolute mean differences with critical range



# Tukey-Kramer Critical Range

$$\text{Critical Range} = Q_U \sqrt{\frac{MSW}{2}\left(\frac{1}{n_j} + \frac{1}{n_{j'}}\right)}$$

where:

$Q_U$ = Value from Studentized Range Distribution with $c$ and $n - c$ degrees of freedom for the desired level of $\alpha$ (see appendix E.9 table)

MSW = Mean Square Within

$n_j$ and $n_{j'}$ = Sample sizes from groups j and j'

# The Tukey-Kramer Procedure: Example

| Club 1 | Club 2 | Club 3 |
|---|---|---|
| 254 | 234 | 200 |
| 263 | 218 | 222 |
| 241 | 235 | 197 |
| 237 | 227 | 206 |
| 251 | 216 | 204 |

1. Compute absolute mean differences:

$$|\bar{x}_1 - \bar{x}_2| = |249.2 - 226.0| = 23.2$$

$$|\bar{x}_1 - \bar{x}_3| = |249.2 - 205.8| = 43.4$$

$$|\bar{x}_2 - \bar{x}_3| = |226.0 - 205.8| = 20.2$$

2. Find the $Q_U$ value from the table in appendix E.10 with $c = 3$ and $(n - c) = (15 - 3) = 12$ degrees of freedom for the desired level of $\alpha$ ($\alpha = 0.05$ used here):

$$\boxed{Q_U = 3.77}$$

# The Tukey-Kramer Procedure: Example

*(continued)*

3. Compute Critical Range:

$$\text{Critical Range} = Q_U\sqrt{\frac{MSW}{2}\left(\frac{1}{n_j}+\frac{1}{n_{j'}}\right)} = 3.77\sqrt{\frac{93.3}{2}\left(\frac{1}{5}+\frac{1}{5}\right)} = 16.285$$

4. Compare:

$$\left|\bar{x}_1 - \bar{x}_2\right| = 23.2$$

$$\left|\bar{x}_1 - \bar{x}_3\right| = 43.4$$

$$\left|\bar{x}_2 - \bar{x}_3\right| = 20.2$$

5. All of the absolute mean differences are greater than critical range. Therefore there is a significant difference between each pair of means at 5% level of significance.

Thus, with 95% confidence we can conclude that the mean distance for club 1 is greater than club 2 and 3, and club 2 is greater than club 3.

# The Randomized Block Design

- Like One-Way ANOVA, we test for equal population means (for different factor levels, for example)...
- ...but we want to control for possible variation from a second factor (with two or more levels)
- Levels of the secondary factor are called blocks

# Partitioning the Variation

- Total variation can now be split into three parts:

$$SST = SSA + SSBL + SSE$$

SST = Total variation
SSA = Among-Group variation
SSBL = Among-Block variation
SSE = Random variation

# Sum of Squares for Blocking

$$SST = SSA + \boxed{SSBL} + SSE$$

$$\boxed{SSBL = c\sum_{i=1}^{r}(\overline{X}_{i.} - \overline{\overline{X}})^2}$$

Where:

c = number of groups

r = number of blocks

$\overline{X}_{i.}$ = mean of all values in block i

$\overline{\overline{X}}$ = grand mean (mean of all data values)

# Partitioning the Variation

- Total variation can now be split into three parts:

$$SST = SSA + SSBL + SSE$$

SST and SSA are computed as they were in One-Way ANOVA

$$SSE = SST - (SSA + SSBL)$$

# Mean Squares

$$\text{MSBL} = \text{Mean square blocking} = \frac{\text{SSBL}}{\text{r} - 1}$$

$$\text{MSA} = \text{Mean square among groups} = \frac{\text{SSA}}{\text{c} - 1}$$

$$MSE = \text{Mean square error} = \frac{SSE}{(r-1)(c-1)}$$

# Randomized Block ANOVA Table

| Source of Variation | SS | df | MS | F ratio |
|---|---|---|---|---|
| Among Treatments | SSA | c - 1 | MSA | $\frac{MSA}{MSE}$ |
| Among Blocks | SSBL | r - 1 | MSBL | $\frac{MSBL}{MSE}$ |
| Error | SSE | (r–1)(c-1) | MSE | |
| Total | SST | rc - 1 | | |

c = number of populations

r = number of blocks

rc = sum of the sample sizes from all populations

df = degrees of freedom

# Blocking Test

$$H_0 : \mu_{1.} = \mu_{2.} = \mu_{3.} = \ldots$$

$$H_1 : \text{Not all block means are equal}$$

$$F = \frac{MSBL}{MSE}$$

- Blocking test: $df_1 = r - 1$
  $df_2 = (r - 1)(c - 1)$

$$\boxed{\text{Reject } H_0 \text{ if } F > F_U}$$

# Main Factor Test

$$H_0 : \mu_{.1} = \mu_{.2} = \mu_{.3} = \ldots = \mu_{.c}$$

$$H_1 : \text{Not all population means are equal}$$

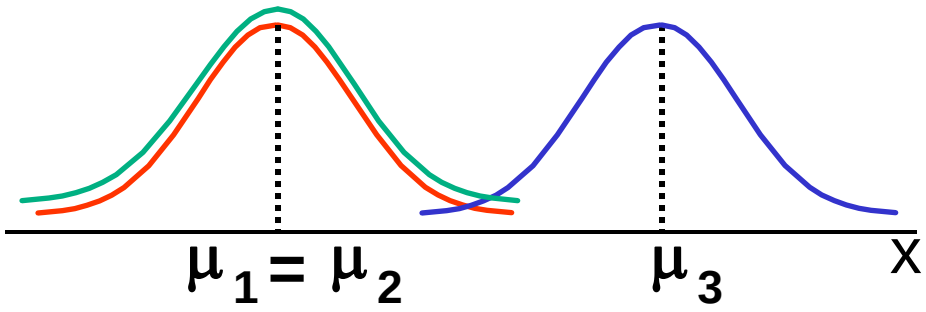
$$F = \frac{MSA}{MSE}$$

- Main Factor test: $df_1 = c - 1$
  $df_2 = (r - 1)(c - 1)$

$$\text{Reject } H_0 \text{ if } F > F_U$$

# The Tukey Procedure

- To test which population means are significantly different
  - e.g.: $\mu_1 = \mu_2 \neq \mu_3$
  - Done after rejection of equal means in randomized block ANOVA design
- Allows pair-wise comparisons
  - Compare absolute mean differences with critical range



# The Tukey Procedure

*(continued)*

$$\text{Critical Range} = Q_u \sqrt{\frac{\text{MSE}}{r}}$$

Compare:

$$\left|\bar{x}_{.1} - \bar{x}_{.2}\right|$$

$$\left|\bar{x}_{.1} - \bar{x}_{.3}\right|$$

$$\left|\bar{x}_{.2} - \bar{x}_{.3}\right|$$

etc...

Is $\left|\bar{x}_{.j} - \bar{x}_{.j'}\right| > \text{Critical Range}$ ?

If the absolute mean difference is greater than the critical range then there is a significant difference between that pair of means at the chosen level of significance.

# Factorial Design: Two-Way ANOVA

- Examines the effect of
  - Two factors of interest on the dependent variable
    - e.g., Percent carbonation and line speed on soft drink bottling process
  - Interaction between the different levels of these two factors
    - e.g., Does the effect of one particular carbonation level depend on which level the line speed is set?

# Two-Way ANOVA

*(continued)*

- Assumptions
  - Populations are normally distributed
  - Populations have equal variances
  - Independent random samples are drawn

# Two-Way ANOVA Sources of Variation

**Two Factors of interest: A and B**

r = number of levels of factor A

c = number of levels of factor B

n′ = number of replications for each cell

n = total number of observations in all cells
(n = rcn′)

$X_{ijk}$ = value of the $k^{th}$ observation of level i of factor A and level j of factor B

# Two-Way ANOVA Sources of Variation

*(continued)*

$$SST = SSA + SSB + SSAB + SSE$$

Degrees of Freedom:

SST
Total Variation

n - 1

SSA
Factor A Variation

r – 1

SSB
Factor B Variation

c – 1

SSAB
Variation due to interaction between A and B

(r – 1)(c – 1)

SSE
Random variation (Error)

rc(n' – 1)

# Two Factor ANOVA Equations

Total Variation:

$$SST = \sum_{i=1}^{r}\sum_{j=1}^{c}\sum_{k=1}^{n'}(X_{ijk} - \bar{\bar{X}})^2$$

Factor A Variation:

$$SSA = cn'\sum_{i=1}^{r}(\bar{X}_{i..} - \bar{\bar{X}})^2$$

Factor B Variation:

$$SSB = rn'\sum_{j=1}^{c}(\bar{X}_{.j.} - \bar{\bar{X}})^2$$

# Two Factor ANOVA Equations

*(continued)*

Interaction Variation:

$$SSAB = n'\sum_{i=1}^{r}\sum_{j=1}^{c}(\bar{X}_{ij.} - \bar{X}_{i..} - \bar{X}_{.j.} + \bar{\bar{X}})^2$$

Sum of Squares Error:

$$SSE = \sum_{i=1}^{r}\sum_{j=1}^{c}\sum_{k=1}^{n'}(X_{ijk} - \bar{X}_{ij.})^2$$

# Two Factor ANOVA Equations

*(continued)*

where:

$$\bar{\bar{X}} = \frac{\sum_{i=1}^{r}\sum_{j=1}^{c}\sum_{k=1}^{n'} X_{ijk}}{rcn'} = \text{Grand Mean}$$

$$\bar{X}_{i..} = \frac{\sum_{j=1}^{c}\sum_{k=1}^{n'} X_{ijk}}{cn'} = \text{Mean of } i^{th} \text{ level of factor A} \quad (i = 1, 2, \ldots, r)$$

$$\bar{X}_{.j.} = \frac{\sum_{i=1}^{r}\sum_{k=1}^{n'} X_{ijk}}{rn'} = \text{Mean of } j^{th} \text{ level of factor B} \quad (j = 1, 2, \ldots, c)$$

$$\bar{X}_{ij.} = \sum_{k=1}^{n'} \frac{X_{ijk}}{n'} = \text{Mean of cell ij}$$

r = number of levels of factor A

c = number of levels of factor B

n′ = number of replications in each cell

# Mean Square Calculations

$$\text{MSA} = \text{Mean square factor A} = \frac{\text{SSA}}{r-1}$$

$$\text{MSB} = \text{Mean square factor B} = \frac{\text{SSB}}{c-1}$$

$$\text{MSAB} = \text{Mean square interaction} = \frac{\text{SSAB}}{(r-1)(c-1)}$$

$$\text{MSE} = \text{Mean square error} = \frac{\text{SSE}}{rc(n'-1)}$$

# Two-Way ANOVA: The F Test Statistic

**F Test for Factor A Effect**

$H_0$: $\mu_{1..} = \mu_{2..} = \mu_{3..} = \cdots$

$H_1$: Not all $\mu_{i..}$ are equal

$$F = \frac{MSA}{MSE}$$

Reject $H_0$ if $F > F_U$

**F Test for Factor B Effect**

$H_0$: $\mu_{.1.} = \mu_{.2.} = \mu_{.3.} = \cdots$

$H_1$: Not all $\mu_{.j.}$ are equal

$$F = \frac{MSB}{MSE}$$

Reject $H_0$ if $F > F_U$

**F Test for Interaction Effect**

$H_0$: the interaction of A and B is equal to zero

$H_1$: interaction of A and B is not zero

$$F = \frac{MSAB}{MSE}$$

Reject $H_0$ if $F > F_U$

# Two-Way ANOVA Summary Table

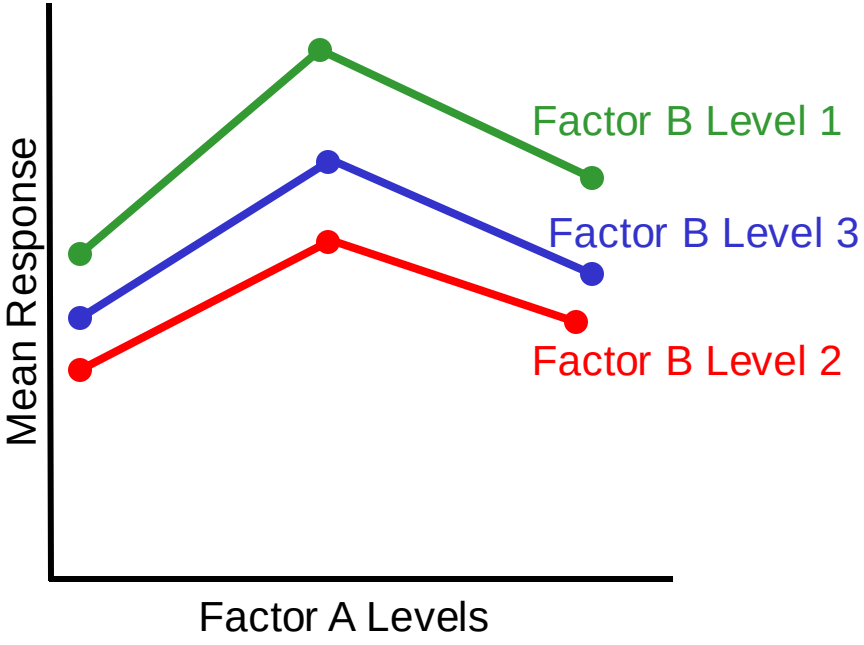
| Source of Variation | Sum of Squares | Degrees of Freedom | Mean Squares | F Statistic |
|---|---|---|---|---|
| **Factor A** | **SSA** | **r – 1** | **MSA** = SSA/(r – 1) | $\frac{\textbf{MSA}}{\textbf{MSE}}$ |
| **Factor B** | **SSB** | **c – 1** | **MSB** = SSB /(c – 1) | $\frac{\textbf{MSB}}{\textbf{MSE}}$ |
| **AB (Interaction)** | **SSAB** | **(r – 1)(c – 1)** | **MSAB** = SSAB / (r – 1)(c – 1) | $\frac{\textbf{MSAB}}{\textbf{MSE}}$ |
| **Error** | **SSE** | **rc(n′ – 1)** | **MSE =** SSE/rc(n’ – 1) | |
| **Total** | **SST** | **n – 1** | | |

# Features of Two-Way ANOVA F Test

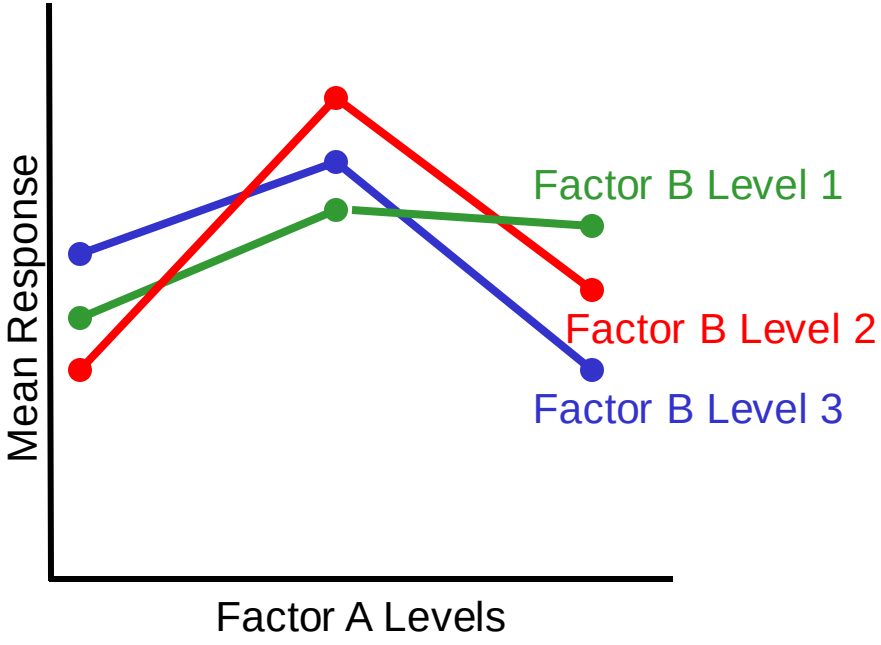
- Degrees of freedom always add up
  - $n-1 = rc(n'-1) + (r-1) + (c-1) + (r-1)(c-1)$
  - Total = error + factor A + factor B + interaction
- The denominator of the F Test is always the same but the numerator is different
- The sums of squares always add up
  - SST = SSE + SSA + SSB + SSAB
  - Total = error + factor A + factor B + interaction

# Examples: Interaction vs. No Interaction

- No interaction:




- Interaction is present:



# Multiple Comparisons: The Tukey Procedure

- Unless there is a significant interaction, you can determine the levels that are significantly different using the Tukey procedure
- Consider all absolute mean differences and compare to the calculated critical range
- Example: Absolute differences for factor A, assuming three factors:

$$\left|\overline{X}_{1..} - \overline{X}_{2..}\right|$$

$$\left|\overline{X}_{1..} - \overline{X}_{3..}\right|$$

$$\left|\overline{X}_{2..} - \overline{X}_{3..}\right|$$

# Multiple Comparisons: The Tukey Procedure

- Critical Range for Factor A:

$$\text{Critical Range} = Q_U \sqrt{\frac{\text{MSE}}{\text{c n'}}}$$

(where $Q_u$ is from Table E.10 with r and rc(n'–1) d.f.)

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- Critical Range for Factor B:

$$\text{Critical Range} = Q_U \sqrt{\frac{\text{MSE}}{\text{r n'}}}$$

(where $Q_u$ is from Table E.10 with c and rc(n'–1) d.f.)

# Chapter Summary

- Described one-way analysis of variance
  - The logic of ANOVA
  - ANOVA assumptions
  - F test for difference in c means
  - The Tukey-Kramer procedure for multiple comparisons
- Considered the Randomized Block Design
  - Treatment and Block Effects
  - Multiple Comparisons: Tukey Procedure
- Described two-way analysis of variance
  - Examined effects of multiple factors
  - Examined interaction between factors